

Chapter 20: Induced Currents and Fields

We will discuss the following:

- **Faraday's Induction Law**
- **Motional emf and Electrical Energy**
- **Lenz Law**
- **Electric Generators**
- **Transformers**

Chapter 20: Induced Currents and Fields

Faraday's Law

Faraday law states, “when the magnetic flux or the magnetic field changes with time, the electromotive force is produced”

The average emf ε induced in a coil of N loops is

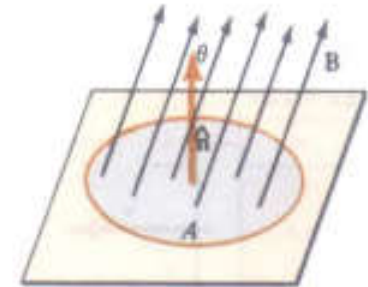
$$\varepsilon = -N \left(\frac{\Phi - \Phi_o}{t - t_o} \right) = -N \frac{\Delta \Phi}{\Delta t}$$

where $\Delta \Phi$ is the change in magnetic flux through one loop and Δt is the time interval during which the change occurs.

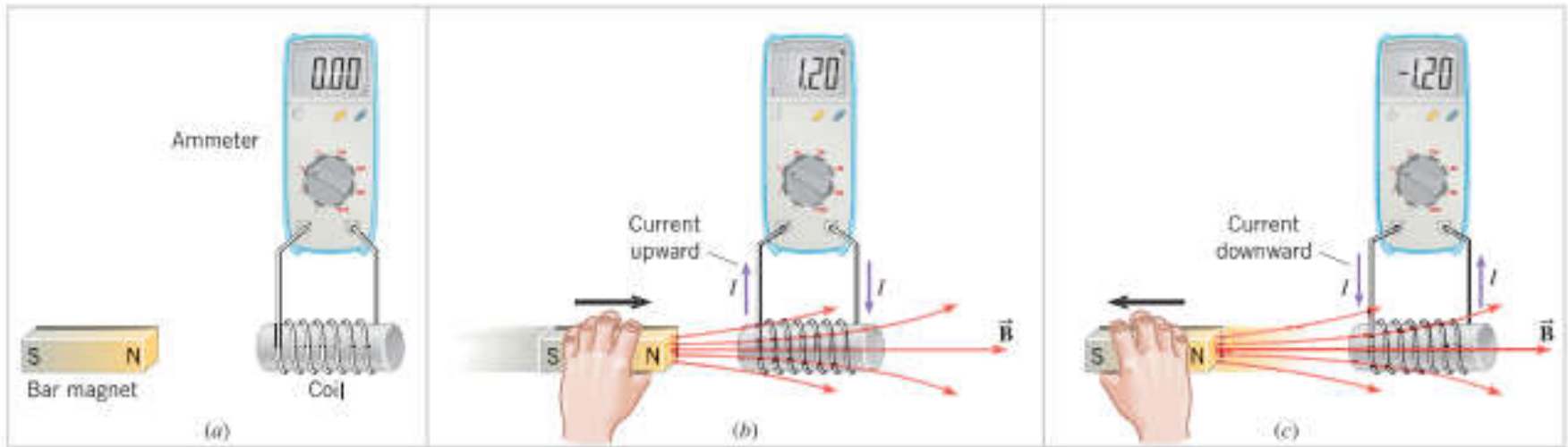
The term $\frac{\Delta \Phi}{\Delta t}$ is the average time rate of change of the flux that passes through one loop.

Magnetic Flux Φ

$$\Phi = BA \cos \theta = \vec{B} \cdot \vec{A}$$



*The magnetic flux Φ through the loop is
 $BA \cos \theta = B_n A$.*



(a) When there is no relative motion between the coil and the bar magnet, there is no current in the coil. (b) A current is created in the coil when the magnet moves toward the coil (c) A current also exists when the magnet moves away from the coil, but the direction of current is opposite to that in (b).

EXAMPLE 20.2: A wire loop of area 0.1m^2 has a resistance of 10Ω (as In Fig.). A magnetic field B normal to loop initially has a magnitude Of 0.2T and is reduced to zero at a uniform rate in 10^{-4}s . Find the emf and the resulting current.

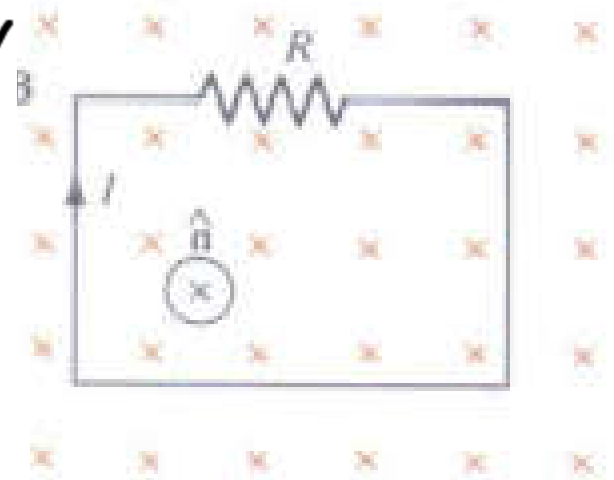
Solution

$$\varepsilon = -\frac{\Delta\Phi}{\Delta t} = -\frac{(0 - BA)}{\Delta t} = \frac{0.1 \times 0.2}{10^{-4}} = 200\text{V}$$

and the induced current is

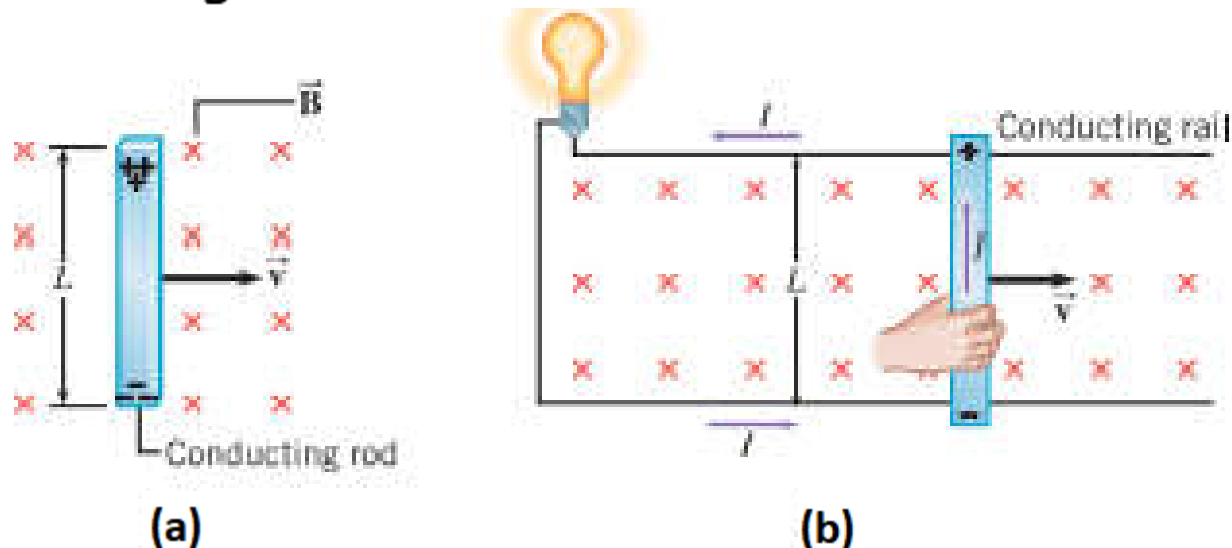
$$I_{\text{ind}} = \frac{\varepsilon}{R} = \frac{200}{10} = 20\text{A}$$

Since I is positive, the current is clockwise direction



■ Motional Emf and Electrical Energy

Motional emf is the emf induced in a conducting rod when it moves in a region of magnetic field. Motional emf is calculated by $\text{emf} = \varepsilon = v B L$. v = speed of the rod, B = magnetic field and L is the length of the rod.



(a) When a conducting rod moves at right angles to a constant field, the magnetic Force causes opposite charges to appear at the ends of the rod, giving rise to an Induced emf. (b) The induced emf causes an induced current I to appear in the circuit.

The change in the magnetic flux as the rod moves to the right is

$$\Delta\Phi = B\Delta A = BLv \Delta t$$

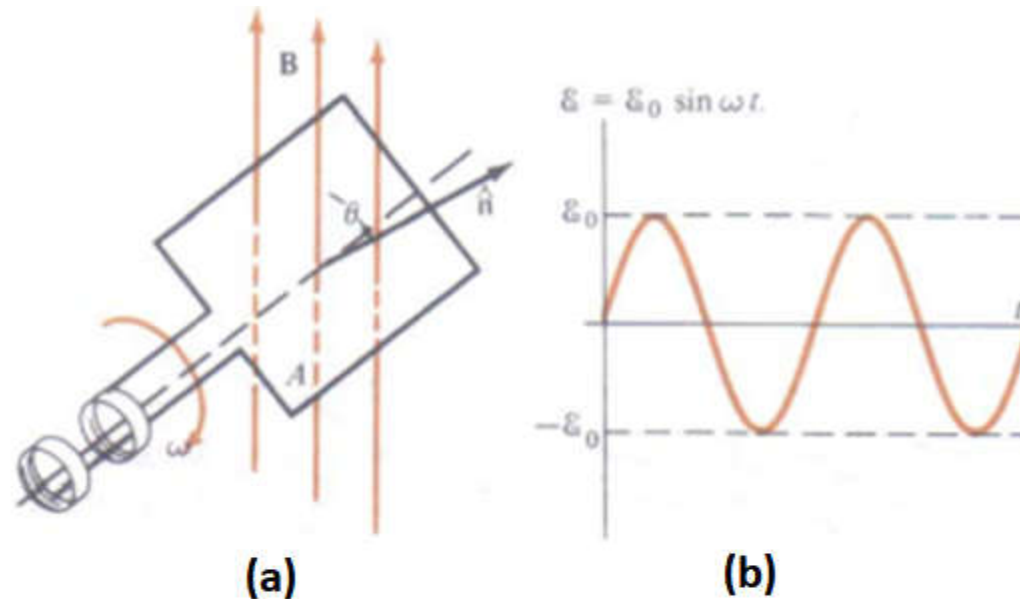
So, the emf is given by

$$\varepsilon = -\frac{\Delta\Phi}{\Delta t} = -BLv$$

The minus sign indicates that the induced current $I = \varepsilon/R$ will be negative, or counterclockwise.

Lenz' law: says that the direction of the induced current will oppose the change in flux that created it.

Electric Generators:

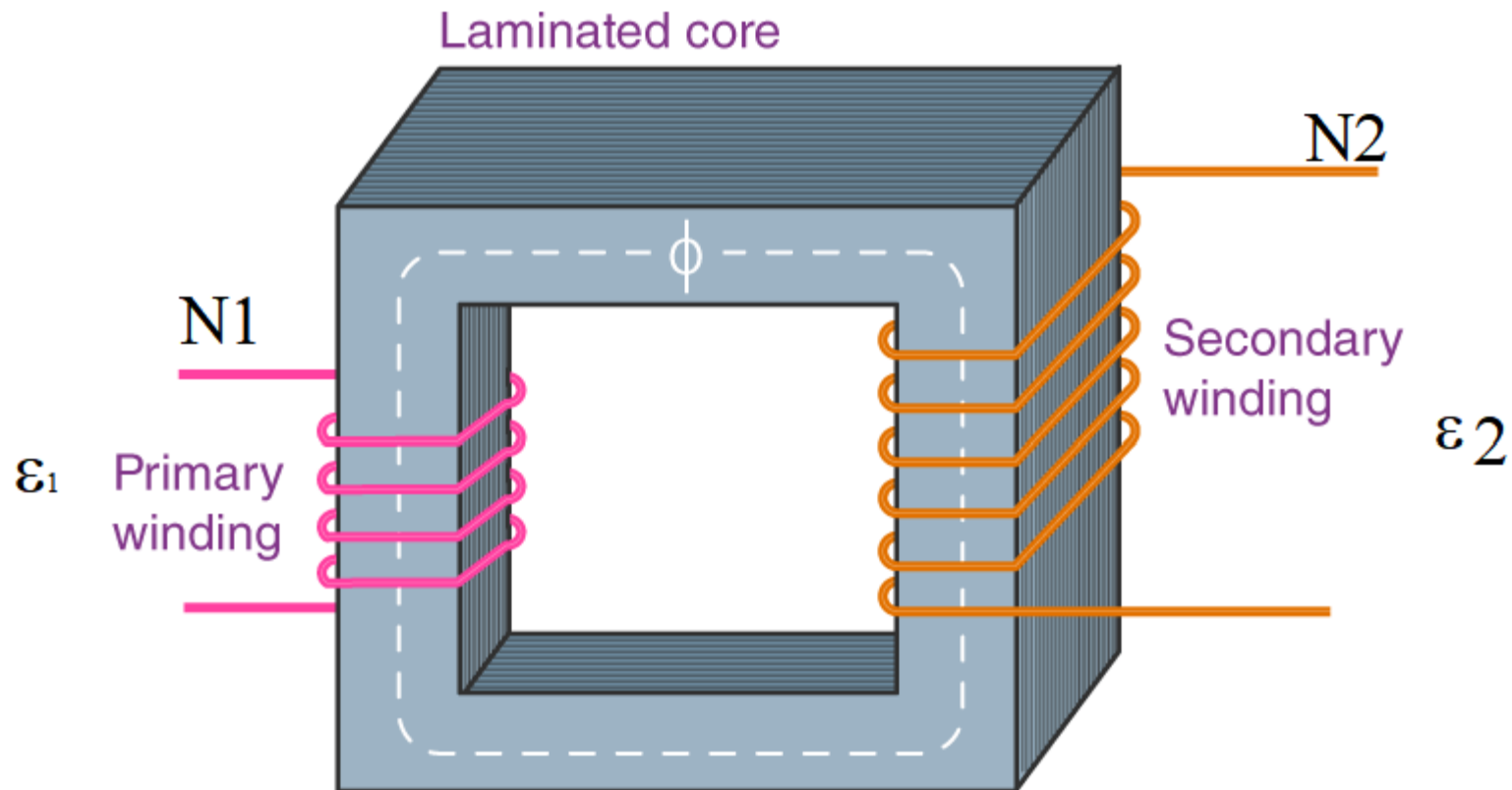


(a) A simplified ac genertor consists of a coil rotating in a magnetic field. (b) The potential difference between the rings

$$\Phi = B_n A = BA \cos(\omega t)$$

$$\mathcal{E} = \left| -\frac{\Delta \Phi}{\Delta t} \right| \Rightarrow \mathcal{E} = \mathcal{E}_0 \sin(\omega t)$$

Transformers:



The relation between ϵ_1 and ϵ_2 is:

$$\epsilon_1 = -N_1 \frac{\Delta \Phi}{\Delta t}$$

$$\epsilon_2 = -N_2 \frac{\Delta \Phi}{\Delta t}$$

$$\Rightarrow \frac{\epsilon_2}{\epsilon_1} = \frac{N_2}{N_1}$$

EXAMPLE 20.4: A transformer reduces the voltage supplied to a Nearby house from 480V ac to 120V. (a) What is the ratio of the turns In the secondary and primary circuits? (b) If the current in primary is 29A, how large is the current in the secondary?

(a) The ratio of turns is

$$\frac{N_2}{N_1} = \frac{\varepsilon_2}{\varepsilon_1} = \frac{120}{480} = \frac{1}{4}$$

(b) $\varepsilon_1 I_1 = \varepsilon_2 I_2$

$$\Rightarrow I_2 = I_1 \frac{\varepsilon_1}{\varepsilon_2} = 20 \times \frac{480}{120} = 80 A$$